

AUTHORIZATION TAXONOMY AUDIT

PART 3 – FINAL AUTHORIZATION TAXONOMY

FAMILY AUT-001: AI-ENHANCED AUTHORIZATION FEATURE MANIPULATION

A. IDENTITY

Field	Value
Attack ID	AUT-001
Attack Family	AI-Enhanced Authorization Feature Manipulation
Attack Subtypes / Variants	Transaction splitting/structuring; Velocity check evasion; Low-and-slow fraud; Threshold-conscious amount optimization; Adaptive timing manipulation
Primary Lifecycle Stage	Authorization (Stage 10)
Secondary / Cross-stage Stages	Payment Initiation (Stage 5 – multiple instructions); AML/Compliance (Stage 17 – structuring detection)
Target / Actor	Authorization risk engines; velocity rules; amount thresholds; floor limits; risk score floors
Attack Objective	Execute large-scale fraudulent transactions where no individual transaction triggers authorization thresholds, enabling sustained fraud extraction over extended periods
Authorization Control Targeted	Per-transaction amount limits; velocity rules (count/time, sum/time); risk-score floors; floor limits for offline authorization

B. FRAUD UNDERSTANDING

What is the attack?

Attackers systematically structure transaction amounts, timing, and velocity to remain below authorization thresholds. GenAI makes this attack *adaptive*—learning from which transactions are approved or declined and adjusting future transaction features to maximize approval rates while maintaining a legitimate-looking profile. The attack does not manipulate the payment instruction itself (that would be Payment Initiation) but rather the *quantitative properties* of the instruction that feed the authorization risk decision.

Traditional mechanism:

Before GenAI, attackers manually split large payments into smaller amounts, distributed transactions across time to avoid velocity rules, and kept amounts below known thresholds. This was static, limited in scale, and detectable by modern ML-based anomaly detection.

Dimension	Pre-GenAI	GenAI-Transformed
Optimization	Manual, static thresholds	Adaptive, learning from approval/decline feedback
Pattern generation	Fixed splitting rules	AI-generated distributions mimicking legitimate cohorts
Scale	Limited, manual	Industrial-scale, automated
Adaptation	None	Real-time adaptation to detection changes

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Dynamically infer thresholds from authorization responses
- Generate amount distributions that mimic legitimate cohort patterns
- Adapt timing and velocity in response to detection feedback
- Operate at industrial scale while maintaining human-like patterns

HARD GenAI Removal Test result: PARTIAL — The attack exists without GenAI, but GenAI transforms it from static, detectable evasion to adaptive, ML-evading optimization.

Prerequisites:

- Access to multiple payment instruments/cards OR a single high-limit card
- Ability to control transaction timing and amounts

- (For GenAI variant) Feedback loop of authorization responses (approve/decline)
- Understanding of target thresholds (discovered through probing)

Attack Flow:

1. Attacker probes authorization system with test transactions across different amounts and timing patterns
2. Observes approval/decline outcomes to infer thresholds (velocity caps, amount limits, risk score floors)
3. GenAI agent develops optimal transaction strategy: amounts just below limits, timing that avoids velocity peaks, patterns that mimic legitimate cohorts
4. Attacker executes fraudulent transactions optimized to stay under thresholds
5. Each transaction individually approved as "low-risk"
6. Cumulative fraud grows over weeks/months
7. If detection changes, GenAI adapts patterns in response

Authorization Lifecycle Point:

Fraud enters at **Feature Generation**—the attacker chooses the amount, timing, and distribution of transactions. The fraud is not in the instruction (Payment Initiation) but in the quantitative properties of the instruction that feed the authorization risk decision.

Signals Being Exploited:

- Transaction amount (kept below thresholds)
- Inter-transaction timing (avoiding velocity peaks)
- Transaction frequency (distributed to stay under velocity caps)
- Merchant selection (low-risk MCCs to avoid risk score increases)

Downstream Payment / Financial Consequence:

- Large fraud volumes executed below per-transaction detection
- Sustained losses over weeks/months before detection
- "High-frequency, low-value 'micro frauds'... can quietly erode up to ~2% of GMV" **【payment-initiation-5.pdf†L4】**
- Delayed detection allows funds to be laundered before intervention

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Account Required?
Transaction	Amounts just below known thresholds (e.g., \$999.99)	Indicates threshold-conscious structuring	Rules (tolerance bands) + Anomaly Detection	No
Transaction	Amount distributions with unnatural consistency	Indicates algorithmically generated amounts	ML (distributional anomaly, entropy)	No
Temporal	Inter-transaction intervals with low entropy	Indicates timing optimization, not human randomness	Sequence Modeling (LSTM/Autoencoder)	No
Temporal	Transaction volume has unnaturally constant mean across time windows	Evading velocity spikes; human behavior has variance	Anomaly Detection (velocity Z-score)	No
Temporal	Gradual monotonic increase in transaction amounts over time	Indicates systematic "probing"	Temporal Trend Analysis	No
Customer/Account	Low variance in daily/monthly spending patterns	Indicates structured, optimized spending	ML anomaly detection	No
Merchant	Consistent use of low-risk MCCs for high-risk purchases	MCC mismatch or gaming	Rules + ML (MCC-amount inconsistency)	No
Network	Similar threshold-just-below patterns across multiple accounts	Indicates coordinated use of the same strategy	Clustering / Graph Analytics	Yes (essential)

Evasion Considerations (Defensive Threat Modeling):

An adaptive attacker might attempt to:

- Introduce random noise (±5% amounts, ±15 min timing) to avoid perfect patterns
- Intersperse legitimate purchases with fraudulent ones
- Mimic seasonal spending patterns (holiday volume increases)
- Dynamically shift to slightly lower thresholds if probing reveals detection
- Use multiple accounts to distribute activity

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Subscription billing	Regular amounts, regular timing	Flagging legitimate Netflix/utility payments
B2B bulk payments	Large amounts near reporting thresholds	Flagging legitimate vendor payments
Employee expense reimbursements	Multiple small transactions	Flagging legitimate business expenses
Seasonal shopping	Velocity bursts	Flagging legitimate holiday shopping
Conscientious budgeting	User deliberately stopping at limits	Flagging legitimate self-controlled spending
Automated payroll	Regular amounts and timing	Flagging legitimate direct deposits

D. SIMULATION INTELLIGENCE

Field	Specification
Simulation Type	Algorithmic (primary) with Hybrid (agentic feedback for adaptation)
Synthetic Customers/Accounts	Payer personas with baseline spending patterns; account velocity profiles (count/hr, sum/hr); varying risk profiles
Synthetic Merchants	MCC-specific amount distribution baselines (e.g., grocery average = \$50, electronics average = \$500); merchant risk tiers
Transactions	Amounts, timestamps, merchant IDs, MCCs, authorization outcomes
Transaction Sequences	Series of 10–1,000 transactions per account per campaign; includes legitimate baseline transactions
Amount Distributions	Sampled just below defined thresholds (e.g., Uniform[\$threshold - X%, threshold]) with periodic noise injection
Timing	Inter-transaction times sampled from distributions that mimic human behavior but optimized to stay under velocity caps
Geography	IP geolocation consistent with customer profile (to avoid geographic flags)
Device/Session	Consistent device fingerprint per account (to appear trusted)
Authorization Decisions	Approve/Decline/Challenge labels for each transaction (simulated risk engine response)
Risk Signals	Per-transaction risk score; velocity flags; threshold breaches; cumulative velocity
Behavioral Evolution	Gradual amount escalation over 30–90 days; pattern mutation after detection events
Network Relationships	Optional—multiple accounts using identical optimization algorithms (for cross-account detection simulation)
GenAI-Generated Behavior	Adaptive policy for amount/timing selection based on reward (approval rate); Bayesian optimization of threshold discovery
Attack Labels	FEATURE_MANIPULATION: SPLITTING ; FEATURE_MANIPULATION: VELOCITY_EVASION ; FEATURE_MANIPULATION: LOW_SLOW ; FEATURE_MANIPULATION: ADAPTIVE
Variant / Mutation Parameters	Manipulation type (amount/timing/velocity); threshold target; noise level; escalation rate; adaptation speed; account diversity
Legitimate Counterpart	Simulated legitimate user spending (no threshold-conscious behavior, natural variance, bursts)
Realistic Edge Cases	Mixed legitimate+fraudulent patterns; partial adaptation failure; high-noise evasion attempts; detection-triggered strategy shift

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI enables attackers to "optimize transaction patterns that evade detection"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
"Modern fraudsters are using AI to generate ... transaction patterns that mimic legitimate buyer behavior"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
"AI now enables adaptive fraud workflows that learn and adapt in real-time"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
"High-frequency, low-value 'micro frauds'... can quietly erode up to ~2% of GMV"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
"Sophisticated attackers evade velocity checks by going 'low and slow' across many accounts and IPs"	Authorization Baseline Research	Research	2025	Industry analysis	VERIFIED

Claim	Source	Publisher	Date	Evidence Type	Confidence
Split transactions occur to stay within a single transaction authorization limit	Authorization Baseline Research	Research	2025	Industry analysis	VERIFIED

F. PRODUCT MAPPING

Generate (Red Team):

- Algorithmic generation of amount distributions that mimic legitimate cohort patterns
- Agentic optimization of timing based on simulated authorization responses
- Parameter variation by threshold type, merchant category, account history
- Mutation: noise injection, gradual escalation, strategy shifts after detection events

Defend (Blue Team):

- Detect amount distributions with unnatural consistency (low variance, threshold-clustering)
- Detect inter-transaction intervals with low entropy
- Detect monotonic amount escalation over time
- Detect velocity that is "too smooth" (unnaturally constant mean)

Detection Approach:

- **Rules:** Tolerance bands around known thresholds
- **Supervised ML:** Classifier on feature vectors (amount relative to threshold, inter-transaction entropy)
- **Anomaly Detection:** Isolation Forest or Autoencoder on transaction feature sequences
- **Sequence Modeling:** LSTM predicting next transaction amount/timing and flagging deviations
- **Cumulative Analytics:** Rolling sum/count monitoring with anomaly detection

Mitigation:

Risk Level	Action	Description
Low	Approve	Normal pattern; no threshold-conscious behavior
Medium	Challenge	Step-up authentication on account; request additional verification
High	Review	Flag for manual/automated review; investigate account pattern
Critical	Hold	Temporarily block transactions from account pending investigation
Confirmed	Decline	Reject transactions; escalate to fraud team

Evaluation:

- Precision (avoiding false positives on legitimate subscription/bulk payments)
- Recall (catching structured patterns)
- Time-to-Detection (should identify within 3–5 transactions)
- False Positive Rate (business friction)

FAMILY AUT-002: ADVERSARIAL FRAUD SCORE MANIPULATION

A. IDENTITY

Field	Value
Attack ID	AUT-002
Attack Family	Adversarial Fraud Score Manipulation
Attack Subtypes / Variants	Model boundary exploration; Feature optimization; Model probing; Black-box surrogate modeling; Decision-boundary exploitation
Primary Lifecycle Stage	Authorization (Stage 10)
Secondary / Cross-stage Stages	Issuer (Stage 11 – model owner); Device/Session (Stage 3 – feature manipulation)
Target / Actor	ML-based fraud scoring models used in authorization; risk score thresholds; model feature encodings
Attack Objective	Manipulate transaction features to reduce fraud scores below approval thresholds, enabling fraudulent transactions to be classified as "low-risk"

Field	Value
Authorization Control Targeted	ML model inference pipelines; risk score thresholds; feature encodings; ensemble scoring logic

B. FRAUD UNDERSTANDING

What is the attack?

Attackers systematically perturb multiple transaction features to manipulate the output of the ML-based fraud scoring model used in authorization—driving the score below the approval threshold without changing the underlying fraudulent intent. Unlike Family 1 (which targets raw thresholds), this attack targets the *model’s learned decision boundary*.

Traditional mechanism:

Before GenAI, attackers manually experimented with feature combinations (amount, MCC, timing, device, location) to find patterns that evaded detection. This was limited, manual, and often detectable because the resulting feature combinations were unusual.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Feature exploration	Manual grid search	Generative optimization of feature vectors
Adaptation	Static experiments	Real-time feedback loop (propose → observe → adapt)
Surrogate modeling	None	Builds surrogate model of target scorer
Scale	Limited probes	Thousands of systematic explorations
Plausibility	Detectable unnatural combinations	Within-distribution plausible vectors

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Systematically explore high-dimensional feature spaces
- Build surrogate models of the target scoring system
- Generate feature combinations that are both low-scoring AND within the legitimate data distribution
- Adapt in real-time to model retraining
- Scale across thousands of accounts with optimized strategies

HARD GenAI Removal Test result: PASS – Without GenAI, the attack is limited and detectable; GenAI enables systematic, adaptive feature-space exploration at scale.

Prerequisites:

- Ability to observe authorization responses (Approved/Declined) for probe transactions
- Control over a wide set of transaction features (amount, MCC, timing, device, location, merchant category)
- Access to multiple accounts for probing without burning a single account

Attack Flow:

1. Attacker identifies target fraud model (e.g., Visa VAA, Mastercard Decision Intelligence, custom issuer model)
2. Submits probe transactions with varied feature vectors, observing approval/decline outcomes
3. GenAI agent builds surrogate model of the target scorer
4. Agent generates optimized feature vectors that minimize predicted fraud score
5. Fraudulent transactions submitted with optimized feature vectors
6. Model scores transactions as "low-risk" → approved
7. If model retrains, agent continues probing and adapting

Authorization Lifecycle Point:

Fraud enters at **Feature Generation**—the attacker manipulates the feature vector before it reaches the ML model. This is not API parameter injection (Gateway) and not instruction manipulation (Payment Initiation)—it is the quantitative values of legitimate features that are subtly shifted to evade the model.

Signals Being Exploited:

- Feature combinations manipulated to minimize model score
- Score threshold—transactions consistently score just below approval threshold
- Model blind spots—features the model has low sensitivity to

Downstream Payment / Financial Consequence:

- Fraudulent transactions approved as "low-risk"
- Fraud losses scale rapidly before detection
- "Active fraud score manipulation" counters "adversarial artificial intelligence learning attacks" 【payment-inititation-5.pdf†L4】
- Model effectiveness degrades without detection

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Account Required?
Transaction	Feature combinations with high "unnaturalness" score	Indicates generated/manipulated vectors	Generative Detector (GAN)	No
Transaction	High density of transactions near the model's decision boundary	Indicates systematic boundary probing	Kernel Density Estimation + Boundary Density Monitoring	No
Account/Behavior	Unnatural feature drift patterns	Indicates model-exploratory behavior	Temporal feature trajectory analysis	No
Authorization History	High proportion of approvals near threshold	Indicates successful evasion	Statistics on approved transactions vs. score distribution	No
Network	Similar "boundary-hugging" patterns across many accounts	Indicates coordinated evasion campaign	Clustering / Graph Analytics	Yes (essential)
Model Drift	Sudden increase in approvals for certain feature combinations	Indicates exploitation of a model blind spot	Monitoring model performance on recent transactions	No

Evasion Considerations (Defensive Threat Modeling):

An adaptive attacker might attempt to:

- Add realistic noise to feature vectors to avoid detection
- Train generative model on real transaction data from the same cohort
- Exploit retraining cycles (attack immediately after model update)
- Modify less-weighted features to shift score without changing high-weight features
- Use multiple accounts to distribute probing

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Legitimate user with variable spending	Feature vectors can be "unusual"	Flagging atypical but genuine behavior
New account with rapid ramp-up	Spending evolves quickly	Flagging legitimate new affluent customers
Business purchasing patterns	Large amounts, consistent merchants	Flagging legitimate corporate spending
Seasonal/event-driven spending	Sudden feature shifts	Flagging legitimate travel/holiday spending

D. SIMULATION INTELLIGENCE

Field	Specification
Simulation Type	Hybrid — Algorithmic for feature vector generation + Agentic for optimization feedback loop
Synthetic Customers/Accounts	Account personas with baseline features; fraud target accounts (synthetic or stolen)
Synthetic Merchants	Merchant profiles with MCC-specific plausible feature distributions
Transactions	Feature vectors: amount, MCC, time, device score, location score, velocity context, authentication strength, prior transaction history
Transaction Sequences	Series of probe transactions (20–200) where features are iteratively adjusted; exploitation phase (fraudulent transactions with optimized vectors)
Amount Distributions	Sampled to explore the decision boundary (not just below thresholds, but across all features)
Timing	Timestamps manipulated to exploit temporal features
Geography	Location coordinates chosen to minimize location-based risk

Field	Specification
Device/Session	Device and session features varied to test model sensitivity
Authorization Decisions	Score output (simulated); Approve/Decline labels
Risk Signals	Feature drift; boundary density; cohort consistency; surrogate model uncertainty
Behavioral Evolution	Feature exploration → exploitation phase; adaptation to retraining cycles
Network Relationships	Multiple accounts sharing the same optimization strategy (for cross-account detection simulation)
GenAI-Generated Behavior	GAN/proxy model for surrogate scoring; RL agent for policy optimization; evolutionary feature search
Attack Labels	ADVERSARIAL_SCORE_MANIPULATION ; MODEL_PROBING ; BOUNDARY_EXPLOITATION
Variant / Mutation Parameters	Model architecture (tree vs. neural); feature set; feedback delay; available probe volume; optimization algorithm
Legitimate Counterpart	Real transaction history from genuine customers (no probe/exploration patterns)
Realistic Edge Cases	Partial model knowledge (white-box vs. black-box); limited probe budget; detection feedback; model retraining disruption

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
"Active fraud score manipulation, to counter ... adversarial artificial intelligence learning attacks"	Payment Initiation Taxonomy	Research	2025	Patent/inference	REASONABLE INFERENCE
"AI-powered fraud workflows that learn and adapt in real-time"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
Visa Deep Authorization uses AI to score transactions based on risk	Visa	Visa	2025	Primary source	VERIFIED
Mastercard Decision Intelligence uses GenAI-driven models	Mastercard	Mastercard	2025	Primary source	VERIFIED
Criminal networks "deploy AI to optimize transaction patterns that evade detection"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE

F. PRODUCT MAPPING

Generate (Red Team):

- GAN/proxy model for surrogate scoring of target fraud model
- RL agent for policy optimization (propose vector → observe score → update)
- Evolutionary feature search to find low-scoring combinations
- Parameter variation by feature subset, optimization algorithm, adaptation speed

Defend (Blue Team):

- Detect generated/manipulated feature vectors
- Detect concentration of transactions near decision boundary
- Detect unnatural feature drift patterns
- Detect coordinated boundary-hugging across accounts

Detection Approach:

- **Adversarial Retraining:** Include adversarial examples in training
- **Boundary Density Monitoring:** Detect transaction concentration near threshold
- **Generative Detector:** GAN trained to distinguish generated vs. real feature vectors
- **Ensemble Diversity:** If multiple models disagree, flag potential adversarial input
- **Surrogate Detection:** Monitor for signs of systematic probing (trial sequences)

Mitigation:

Risk Level	Action	Description
Low	Approve	Normal pattern; no adversarial indicators
Medium	Challenge	Step-up authentication if feature vector flagged as generated
High	Review	Manual review of account behavior; investigate feature patterns
Critical	Decline	Confirmed adversarial pattern; escalate to fraud team; consider model retraining

Evaluation:

- Precision (avoiding false positives on genuine unusual behavior)
- Recall (catching adversarial feature vectors)
- False Positive Rate
- Detection latency (should be real-time)

FAMILY AUT-003: AI-COORDINATED DISTRIBUTED FRAUD

A. IDENTITY

Field	Value
Attack ID	AUT-003
Attack Family	AI-Coordinated Distributed Fraud
Attack Subtypes / Variants	Multi-account coordinated fraud; Synthetic account farming; Parallel fraud operations; Coordinated bust-out networks; Fraud mesh orchestration
Primary Lifecycle Stage	Authorization (Stage 10) — per-account decisions
Secondary / Cross-stage Stages	Account Creation (Stage 2 — synthetic accounts); Payment Initiation (Stage 5 — multiple instructions); AML/Compliance (Stage 17 — network detection)
Target / Actor	Per-account fraud models; entity resolution; velocity rules (per-account); merchant concentration limits; graph-based detection
Attack Objective	Execute industrial-scale fraud across hundreds/thousands of accounts, ensuring no single account exceeds detection thresholds, enabling massive fraud extraction while each account appears individually legitimate
Authorization Control Targeted	Per-account fraud scoring; entity resolution; velocity aggregation; merchant concentration monitoring

B. FRAUD UNDERSTANDING

What is the attack?

Attackers orchestrate fraudulent transactions across hundreds or thousands of accounts (synthetic or compromised), ensuring no single account exceeds velocity, amount, or risk thresholds. The aggregate fraud is massive, but each account appears individually legitimate. GenAI coordinates the timing, sequencing, and amounts across accounts to avoid detection by entity-resolution and graph-based analytics.

"Fraud today isn't a single stolen card—networks of mule accounts, synthetic identities, complicit merchants, and cross-border facilitators work together in orchestrated rings." [【acquirer-7.pdf†L10】](#)

Traditional mechanism:

Before GenAI, distributed fraud required manual coordination across accounts, which was limited in scale and detectable by graph analytics. Each account required individual management; coordination was human-limited.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Scale	Dozens of accounts	Hundreds/thousands of accounts
Coordination	Manual, limited	AI-agent orchestrated, autonomous
Adaptation	None	Real-time adaptation to detection
Patterning	Detectable uniformity	Individualized patterns per account
Infrastructure	Static	Dynamically rotated

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Generate thousands of credible synthetic account personas
- Coordinate transaction timing across thousands of accounts
- Adapt orchestration patterns in real-time to avoid detection
- Create individualized transaction patterns for each account
- Rotate infrastructure dynamically

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability: industrial-scale, adaptive orchestration of distributed fraud.

Prerequisites:

- Large inventory of synthetic or compromised accounts
- Infrastructure to manage many accounts
- GenAI coordination agent to optimize collective behavior
- Understanding of entity resolution and graph detection thresholds

Attack Flow:

1. Attacker creates/generates thousands of synthetic accounts (or compromises existing accounts)
2. GenAI agent assigns each account a unique behavioral profile (spending patterns, merchants, amounts, timing)
3. Agent orchestrates transaction timing across accounts to avoid aggregate velocity peaks
4. Each account transacts individually below its per-account detection thresholds
5. Graph analytics fails to link accounts (minimal shared attributes, phase-shifted lifecycles)
6. Cumulative fraud volume is substantial
7. If a subset is detected, agent isolates and reconfigures remaining accounts

Authorization Lifecycle Point:

Fraud enters at **Transaction Initiation for each account**—the fraud is in the collective orchestration of these initiations, not in any single instruction. The authorization decision for each transaction is normal; the attack targets the *inability of per-account systems to see the whole picture*.

Signals Being Exploited:

- Per-account transaction counts (each stays under threshold)
- Per-account amounts (each stays under threshold)
- Aggregate detection (entity resolution, graph analytics)

Downstream Payment / Financial Consequence:

- Industrial-scale fraud executed below per-account detection
- "3,600+ scam merchant accounts identified in 2025, up 2.5x from 2024" **【acquirer-7.pdf†L10】**
- "Network of 71 purchase scam websites linked to 12 shared merchant accounts" **【acquirer-7.pdf†L12】**
- "450%+ surge in dark-web posts discussing 'AI Agent' tools" for fraud **【acquirer-7.pdf†L10】**

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Account Required?
Network/Graph	Accounts sharing infrastructure (IPs, device fingerprints, email domains)	Indicates coordinated campaign	Graph Neural Network; Entity Resolution	Yes (essential)
Network/Graph	Accounts with similar temporal lifecycle patterns	Indicates orchestrated farming	Temporal Clustering on account lifecycles	Yes (essential)
Network/Graph	Anti-correlated activity timing (one pauses, another starts)	Indicates deliberate coordination	Time-series cross-correlation analysis	Yes (essential)
Network/Graph	Unusual diversity of behavior across accounts (too heterogeneous)	May indicate synthetic generation	Anomaly detection on portfolio entropy	Yes
Temporal	Suspicious phase-shifted patterns (identical lifecycles, different start dates)	Indicates batch creation	Phase analysis on lifecycle features	Yes (essential)
Temporal	Coordinated dormancy and reactivation patterns	Indicates orchestrated account farming	Temporal pattern analysis	Yes
Merchant	High number of accounts transacting with the same low-risk MCCs in similar amounts	Indicates coordinated laundering	Merchant concentration + account diversity analysis	Yes
Authorization History	All accounts have similar approval rates or decline patterns	Indicates uniform optimization strategy	Statistical comparison of approval rates across accounts	Yes
Customer/Account	Account creation patterns (batch creation, similar attributes)	Indicates synthetic account farming	Entity resolution; creation pattern analysis	Yes

Evasion Considerations (Defensive Threat Modeling):

An adaptive attacker might attempt to:

- Rotate infrastructure at the account level (unique IPs, devices, email domains)
- Add natural variance to lifecycles (random dormancy, varying ramp-ups)

- Inject "noise" accounts (purely legitimate) to obscure the cluster
- Stay just below graph detection thresholds
- Dynamically adapt to discovery (isolate detected sub-clusters)
- Use decoy accounts to confuse entity resolution

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Payment facilitator (PayFac)	Many sub-merchants under one aggregator	Flagging legitimate merchant portfolios
Multi-location business	Multiple accounts for different locations	Flagging legitimate franchise operations
Employee benefits platform	Multiple accounts with similar activity	Flagging legitimate corporate benefits
Family/business group	Related accounts with similar behavior	Flagging legitimate family accounts
A/B testing by a merchant	Slightly different behavior patterns across accounts	Flagging legitimate marketing experiments

D. SIMULATION INTELLIGENCE

Field	Specification
Simulation Type	Hybrid — Algorithmic for per-account transaction generation + Agentic for network-wide coordination orchestration
Synthetic Customers/Accounts	Hundreds/thousands of synthetic personas with coherent demographics, account histories, and behavioral baselines; varying risk profiles
Synthetic Merchants	A portfolio of merchants (some legitimate, some laundering nodes) across various MCCs
Transactions	Per-account transaction sequences; across-network transaction schedules; varying amounts and MCCs
Transaction Sequences	Orchestrated to avoid aggregate velocity peaks (e.g., total volume per hour across all accounts stays below platform alert levels)
Amount Distributions	Diversified across accounts; some random, some structured, to mimic natural heterogeneity
Timing	Phase-shifted patterns; anti-correlated activity to avoid concentration; coordinated dormancy/reactivation
Geography	IP and location rotated across accounts to avoid geographic clustering
Device/Session	Unique device fingerprints per account; generated by GenAI (see DFS-001)
Authorization Decisions	Per-account approve/decline patterns; aggregate approval rate across the network
Risk Signals	Graph connectivity scores; shared-feature density; lifecycle phase correlations; portfolio entropy
Behavioral Evolution	Account lifecycles: creation → dormancy → ramp-up → high activity → bust-out or fade-out
Network Relationships	Deliberately minimized shared attributes; decoy accounts to confuse graph detection; hidden coordination graph
GenAI-Generated Behavior	Orchestration agent scheduling transactions across accounts; anti-correlated timing; adaptation to detection feedback; infrastructure rotation
Attack Labels	COORDINATED_DISTRIBUTED_FRAUD ; ACCOUNT_FARMING ; PARALLEL_FRAUD ; FRAUD_MESH
Variant / Mutation Parameters	Number of accounts; coordination intensity; diversification level; infrastructure rotation frequency; graph avoidance strength
Legitimate Counterpart	Simulated legitimate multi-account networks (e.g., payment facilitators, corporate groups) with no orchestration
Realistic Edge Cases	Mixed legitimate+fradulent accounts; detection evasion with decoy accounts; partial account compromise; sub-cluster isolation

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
"3,600+ scam merchant accounts identified in 2025, up 2.5x from 2024"	Acquirer Taxonomy	Research	2025	Threat intelligence	VERIFIED
"Network of 71 purchase scam websites linked to 12 shared merchant accounts"	Acquirer Taxonomy	Research	2025	Threat intelligence	VERIFIED
"450%+ surge in dark-web posts discussing 'AI Agent' tools"	Visa PERC	Visa	2025	Payment network	VERIFIED
"Criminal networks use generative AI for parallel operations"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
"AI now enables adaptive fraud workflows that learn and adapt in real-time"	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE
"Fraud actors using agentic techniques to make malicious activity appear more natural, adaptive, and persistent"	MRC	MRC	2025	Industry analysis	VERIFIED
"Fraud today isn't a single stolen card—networks of mule accounts, synthetic identities, complicit merchants work together"	TigerGraph	TigerGraph	2025	Industry report	VERIFIED

F. PRODUCT MAPPING

Generate (Red Team):

- Algorithmic generation of synthetic account personas and transaction histories
- Agentic orchestration of transaction timing across accounts
- Generation of phase-shifted lifecycle patterns
- Parameter variation by number of accounts, coordination intensity, infrastructure rotation
- Mutation: adaptation to detection, isolation of detected sub-clusters

Defend (Blue Team):

- Detect coordinated account lifecycle patterns
- Detect anti-correlated timing across accounts
- Detect shared infrastructure (even if varied)
- Detect unnatural homogeneity in approval rates
- Detect synthetic account creation patterns

Detection Approach:

- **Graph Neural Network:** Detect hidden linkages via graph community detection
- **Temporal Clustering:** Group accounts by lifecycle phase alignment
- **Anomaly Detection on Portfolio Metrics:** Monitor portfolio-level entropy and velocity distribution
- **Cross-correlation Analysis:** Flag phase-shifted identical patterns
- **Entity Resolution:** Link accounts by shared infrastructure

Mitigation:

Risk Level	Action	Description
Low	Approve	Low graph-risk; normal portfolio patterns
Medium	Challenge	Step-up authentication on accounts with network similarity
High	Review	Network-level investigation; analyze account cohorts
Critical	Hold	Temporarily block activity from suspicious account cohorts
Confirmed	Decline	Close all linked accounts; escalate to fraud team; share intelligence

Evaluation:

- Precision (avoiding false positives on legitimate PayFac/corporate groups)
- Recall (catching coordinated networks)
- Time-to-Detection (should identify pattern within 3-5 lifecycle phases)
- False Positive Rate

PART 4 – FINAL COVERAGE AUDIT

Authorization Attack Surface	Considered?	Covered By	Separate Family?	Reason
Authorization-model evasion	✓	AUT-001, AUT-002, AUT-003	No	Covered across all three families
Fraud-score evasion	✓	AUT-002	Yes	Distinct mechanism: targets ML model
Velocity manipulation	✓	AUT-001	No	Variant of Feature Manipulation
Threshold avoidance	✓	AUT-001	No	Variant of Feature Manipulation

Authorization Attack Surface	Considered?	Covered By	Separate Family?	Reason
Low-and-slow behavior	✓	AUT-001	No	Variant of Feature Manipulation
Behavioral mimicry	MOVED → Device/Session (BBE-001)	—	No	Already covered in Device/Session taxonomy 【device-session-3.pdf†L15】
Transaction-feature manipulation	✓	AUT-001, AUT-002	No	Mechanism across families
Amount manipulation	✓	AUT-001	No	Variant of Feature Manipulation
Timing manipulation	✓	AUT-001	No	Variant of Feature Manipulation
Merchant-selection manipulation	MOVED → Acquirer (ACQ-004)	—	No	Already covered in Acquirer taxonomy 【acquirer-7.pdf†L13】
Geography manipulation	✓	AUT-001, AUT-002	No	Feature manipulation variant
Transaction-sequence manipulation	✓	AUT-001, AUT-003	No	Covered across families
Rule-based control evasion	✓	AUT-001	No	Feature Manipulation covers rules
Distributed transaction behavior	✓	AUT-003	Yes	Distinct: cross-account orchestration
Adversarial ML/model manipulation	✓	AUT-002	Yes	Distinct: model decision-boundary targeting
Model probing	✓	AUT-002	No	Variant of Adversarial Score Manipulation
Model behavior inference	✓	AUT-002	No	Variant of Adversarial Score Manipulation
Decision-boundary exploitation	✓	AUT-002	No	Variant of Adversarial Score Manipulation
Coordinated multi-account attacks	✓	AUT-003	Yes	Distinct: industrial-scale orchestration
Coordinated multi-merchant attacks	✓	AUT-003	No	Variant of Distributed Fraud
Adaptive GenAI transaction behavior	✓	AUT-001, AUT-002, AUT-003	No	Capability across all three families
Agentic adaptation to authorization outcomes	✓	AUT-001, AUT-002, AUT-003	No	Capability across all three families
Transaction splitting	✓	AUT-001	No	Variant of Feature Manipulation
Transaction structuring	✓	AUT-001	No	Variant of Feature Manipulation
Floor limit bypass	DROPPED	—	No	No GenAI role; legacy control
Offline authorization abuse	DROPPED	—	No	No GenAI role
Pre-play attack (EMV)	DROPPED	—	No	No GenAI role; cryptographic flaw
Forced authorization	DROPPED	—	No	No GenAI role; physical workflow flaw
Collusion fraud	DROPPED	—	No	No GenAI role; fraud-of-intent
Timing-based evasion	DROPPED	—	No	No GenAI role; simple scheduling
MCC manipulation	MOVED → Acquirer (ACQ-004)	—	No	Already covered 【acquirer-7.pdf†L13】
Transaction amount manipulation	MOVED → Gateway/Processor (GP-001)	—	No	Already covered 【gateway-processor-08.pdf†L1】
Card testing	MOVED → Gateway/Processor (GP-003)	—	No	Already covered 【gateway-processor-08.pdf†L6】
Behavioral mimicry	MOVED → Device/Session (BBE-001)	—	No	Already covered 【device-session-3.pdf†L15】

PART 5 – FINAL SUMMARY

Table 1: FINAL AUTHORIZATION ATTACK TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Authorization Control	Reason for Distinctness
AUT-001	AI-Enhanced Authorization Feature Manipulation	Transaction splitting; Velocity evasion; Low-and-slow fraud; Threshold optimization	PARTIAL	Algorithmic + Agentic	Authorization	Amount limits; Velocity rules; Risk floors	GenAI enables adaptive, ML-evading optimization of transaction features—qualitative leap from static threshold evasion
AUT-002	Adversarial Fraud Score Manipulation	Model boundary exploration; Feature optimization; Model probing; Surrogate modeling	PASS	Hybrid	Authorization	ML model inference; Score thresholds	GenAI enables systematic, adaptive feature-space exploration to evade ML models—qualitative leap from manual probing
AUT-003	AI-Coordinated Distributed Fraud	Multi-account coordination; Synthetic account farming; Parallel fraud operations	PASS	Hybrid	Authorization	Per-account detection; Entity resolution	GenAI enables industrial-scale autonomous orchestration—qualitative leap from manual coordination

Table 2: ATTACK × OBSERVABLE SIGNAL

Attack Family	Transaction	Customer/Account	Merchant	Device	Session	Behavioral	Temporal	Geographic	Network
AUT-001: Feature Manipulation	Amounts just-below thresholds; Low variance; Structured patterns	Unnatural spending consistency; Low velocity variance	Consistent low-risk MCC use	Consistent (trusted)	Normal	N/A	Inter-transaction low entropy; Velocity smoothing	Consistent with profile	Similar threshold patterns (optional)
AUT-002: Adversarial Score Manipulation	Feature vectors with high unnaturalness; Boundary density	Feature drift; Model-exploratory patterns	MCC-feature inconsistencies	Feature manipulation	Normal	N/A	Probe→exploit patterns	Feature manipulation	Boundary-hugging patterns
AUT-003: Distributed Fraud	Diversified across accounts	Phase-shifted lifecycles; Synthetic attributes	Diversified MCCs	Unique per account; Rotated	Normal	N/A	Anti-correlated timing; Synchronized patterns	Rotated	Shared infrastructure Decoy patterns

Table 3: ATTACK × SIMULATION

Attack Family	Synthetic Customers	Synthetic Accounts	Synthetic Merchants	Transactions	Sequences	Timing	Geography	Risk Signals	Agentic
AUT-001: Feature Manipulation	Payer personas with baselines	Single account per campaign	MCC-specific baselines	Amounts; Timestamps; MCCs	10–1,000 per account	Optimized inter-transit intervals; Velocity smoothing	Consistent with profile	Threshold breaches; Velocity flags; Score floors	Algorithm
AUT-002: Adversarial Score Manipulation	Account personas with baselines	Target accounts	Merchant profiles	Full feature vectors	20–200 probes per account	Varying; Optimized per vector	Manipulated	Model score; Boundary density; Feature drift	Hybrid
AUT-003: Distributed Fraud	Hundreds/thousands of synthetic personas	Hundreds/thousands of accounts	Portfolio of merchants	Diversified across accounts	Orchestrated schedules	Phase-shifted; Anti-correlated	Rotated	Graph connectivity; Lifecycle correlations; Portfolio entropy	Hybrid

Table 4: TAXONOMY DECISIONS

Candidate	Decision	Reason
Transaction Splitting	MERGE → AUT-001	Same mechanism: adaptive feature manipulation to stay below thresholds
Velocity Check Evasion	MERGE → AUT-001	Same mechanism: adaptive pacing to avoid detection

Candidate	Decision	Reason
Low-and-Slow Fraud	MERGE → AUT-001	Same mechanism: systematic threshold-conscious behavior
Adversarial Fraud Score Manipulation	KEEP (AUT-002)	Distinct: targets ML model decision boundaries, not raw thresholds
Distributed/Coordinated Fraud	KEEP (AUT-003)	Distinct: cross-account orchestration requiring graph/network detection
Model Probing / Inference	MERGE → AUT-002	Variant of adversarial score manipulation
Decision-Boundary Exploitation	MERGE → AUT-002	Variant of adversarial score manipulation
Multi-Merchant Coordination	MERGE → AUT-003	Variant of distributed fraud
MCC Manipulation	MOVE → Acquirer (ACQ-004)	Already covered in Acquirer taxonomy 【acquirer-7.pdf†L13】
Transaction Amount Manipulation	MOVE → Gateway/Processor (GP-001)	Already covered 【gateway-processor-08.pdf†L1】
Card Testing	MOVE → Gateway/Processor (GP-003)	Already covered 【gateway-processor-08.pdf†L6】
Behavioral Mimicry	MOVE → Device/Session (BBE-001)	Already covered 【device-session-3.pdf†L15】
Floor Limit Bypass	DROP	No GenAI role; attack works identically without AI
Pre-Play Attack (EMV)	DROP	No GenAI role; cryptographic protocol flaw
Forced Authorization	DROP	No GenAI role; physical terminal workflow flaw
Collusion Fraud	DROP	No GenAI role; fraud-of-intent
Timing-Based Evasion	DROP	No GenAI role; simple scheduling

FINAL METRICS

Metric	Value
Total final attack families	3
Total variants	10+ (across 3 families)
Important candidates merged	Transaction Splitting → AUT-001; Velocity Evasion → AUT-001; Low-and-Slow → AUT-001; Model Probing → AUT-002; Decision-Boundary Exploitation → AUT-002; Multi-Merchant Coordination → AUT-003
Important candidates moved	MCC Manipulation → Acquirer (ACQ-004); Transaction Amount Manipulation → Gateway/Processor (GP-001); Card Testing → Gateway/Processor (GP-003); Behavioral Mimicry → Device/Session (BBE-001)
Important candidates dropped	Floor Limit Bypass (no GenAI role); Pre-Play Attack (no GenAI role); Forced Authorization (no GenAI role); Collusion Fraud (no GenAI role); Timing-Based Evasion (no GenAI role)
Theoretical/future threats	Agentic self-learning fraud models; Real-time adversarial model poisoning; Cross-rail coordinated attacks; GenAI-generated behavioral profiles that defeat graph detection
Remaining research gaps	Real-world frequency of adversarial score manipulation in production; Specific detection evasion effectiveness against Visa VAA/Mastercard Decision Intelligence; Effectiveness of cross-consortium graph intelligence against AI-coordinated networks
Authorization control surfaces covered	Amount thresholds; Velocity rules; Risk score floors; ML model inference; Score thresholds; Per-account detection; Entity resolution; Merchant concentration
Important Authorization surfaces not represented as separate families	Behavioral mimicry (moved to Device/Session); MCC manipulation (moved to Acquirer); Card testing (moved to Gateway/Processor); Transaction amount manipulation (moved to Gateway/Processor); Floor limit bypass (dropped—no GenAI role)

UNIFYING THEME

The three Authorization families share a single meta-mechanism: making fraudulent transactions look indistinguishable from legitimate ones to the authorization system, but they operate at different levels. AUT-001 manipulates raw thresholds, AUT-002 manipulates the ML model's internal representation, and AUT-003 manipulates the system's view of the collective. Without GenAI, these attacks are limited, manual, and detectable. With GenAI, they become adaptive, industrial-scale, and capable of evading even sophisticated ML defenses.

GenAI transforms authorization fraud from "can we fool the system?" to "can we generate an entirely indistinguishable transaction reality that the authorization system accepts as legitimate?"

FINAL TAXONOMY COMPARISON

Stage	Number of Families

Stage	Number of Families
Identity/KYC	5
Account Creation	3
Device/Session	5
Authentication	4
Payment Initiation	4
Merchant	6
Acquirer	4
Gateway/Processor	7
Payment Rail	1
Authorization	3
TOTAL	42 families

The Authorization taxonomy has **3 families**, emerging naturally from evidence and the HARD GenAI Removal Test—not forced. This is consistent with the principle that Authorization is a narrower lifecycle stage than Identity/KYC or Device/Session, focused specifically on the real-time risk decision.

END OF FINAL AUTHORIZATION MASTER FRAUD INTELLIGENCE DOCUMENT